

Oracle on Oracle

Objectives

After completing this lesson, you should be able to:

- Prepare your Oracle Linux server for Oracle Database installation
- Use Oracle Pre-Install RPM
- Create Oracle software user and group accounts
- Set kernel parameters for Oracle Database
- Set Oracle database shell limits
- Configure HugePages
- Configure Oracle Database Smart Flash Cache (DBSFC)
- Describe Oracle ASM, ASM Library Driver (ASMLib), and ASM Filter Driver (ASMFD)



Oracle Pre-Install RPM

Oracle Database Pre-Install RPM for Oracle Linux:

- Completes most pre-installation configuration tasks
- Downloads and installs various software packages and specific versions needed for database installation
- Creates the user `oracle` and the groups `oinstall` and `dba`
- Updates kernel parameters
 - These reside in `/etc/sysctl.d/99-oracle-database-server-18c-preinstall-sysctl.conf`
- Sets hard and soft shell resource limits in the `/etc/security/limits.d` directory
- Sets `numa=off` in the kernel boot parameters for x86_64 machines
- Sets `transparent_hugepage=never` in the kernel boot parameters

Oracle Software User Accounts

- The Oracle database software owner:
 - Is commonly named `oracle`
 - Runs the OUI and has full privileges to install, uninstall, and patch Oracle software
 - Cannot be `root`
- The owner of the `httpd` process is:
 - A low-privileged OS user
 - Usually provided by the `nobody` user
- Database operations require a few more users:
 - Members of `OSOPER` group can start, stop, back up, and recover the database.
 - Members of the `OSDBA` group have `OSOPER` privileges, can create and drop a database, and create other `OSDBA` members.



Oracle Automatic Storage Management (ASM) Groups

- The `OSASM` group for Oracle ASM Administration:
 - This is commonly named `asmadmin`.
 - Create this group as a separate group if you want to have separate administration privileges groups for Oracle ASM and Oracle Database administrators.
- The `OSDBA` group for Oracle ASM:
 - This is commonly named `asmdba`.
 - The Oracle Grid Infrastructure software owner (typically, `grid`) must be a member.
 - Membership in this group enables access to the files managed by Oracle ASM.
- The `OSOPER` group for Oracle ASM:
 - This is an optional group.
 - This is commonly named `asmoper`.
 - Create this group if you want a separate group of operating system users to have a limited set of Oracle instance administrative privileges.

For example:

```
# usermod -aG asmadmin oracle  
# usermod -aG asmdba oracle
```

System Resource Tuning

- An Oracle database instance requires certain system resources.
- Shared memory must be adjusted for database use.
- Shared memory system uses semaphores, which must be adjusted.
- Each dedicated server process requires a network port.
- Larger network buffers are recommended.
- The maximum number of open files per process must be increased.
- Shell limit settings must be increased.

Linux Shared Memory

- Three shared memory-related kernel parameters:
 - `SHMMNI`: The maximum number of system-wide shared memory segments
 - `SHMMAX`: The maximum size of each segment
 - `SHMALL`: The maximum number of shared memory pages system wide
- For Oracle database, set `SHMMAX` $\geq$ the largest SGA.
- Set shared memory kernel parameters in: `/etc/sysctl.d/97-oracle-database-sysctl.conf`
- Parameters are viewable in:

```
# ls /proc/sys/kernel/sh*  
shmall      shmmx      shmmni
```

Semaphores

- Semaphores are a method of controlling access to critical resources.
- The Oracle instance uses semaphores to control access to shared memory.
- Semaphores are allocated based on the `PROCESSES` initialization parameter.
- All four semaphore parameters are set by a single `kernel.sem` parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:
 - `semmsl`: Maximum number of semaphores per set
 - `semmns`: Total number of semaphores in the system
 - `semopm`: Maximum number of operations per `semop` call
 - `semmni`: Maximum number of semaphore sets

Minimum configurations:

Setting in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:

`kernel.sem = 250 32000 100 128`

- For **`semms1`**: 250

- Maximum number of semaphores per set
- or the largest PROCESSES parameter of an Oracle database plus 50

- For **`semmns`**: 32000

- or sum of the PROCESSES parameters for each Oracle database, and adding an additional 25 to 50 for each database

- For **`semopm`**: 100

- Maximum number of operations per semop call

- For **`semmni`**: 128

- Maximum number of semaphore sets

Network Tuning

- Socket parameters:
 - An IP port is assigned to a server process when it starts.
 - An IP port is used to communicate with the user process.
 - The default range is 32768 through 61000. Change it if not adequate. Example:

```
# cat /proc/sys/net/ipv4/ip_local_port_range
9000    65500
```

- TCP/IP window size parameters:
 - Define read (`rmem`) and write (`wmem`) window sizes.
 - Set the default and maximum memory allocated for the network send and receive buffers.

```
# ls /proc/sys/net/core/[rw]mem*
rmem_default      wmem_default
rmem_max          wmem_max
```

Network Tuning - Recommendation

- Change the socket receive buffer:

```
# sysctl -w net.core.rmem_default=262144
```

- Change the socket send buffer:

```
# sysctl -w net.core.wmem_default=262144
```

- Change the maximum socket receive buffer size:

```
# sysctl -w net.core.rmem_max=262144
```

- Change the maximum socket send buffer size

```
# sysctl -w net.core.wmem_default=262144
```

Setting the File Handles Parameter

- The File Handles parameter (`fs.file-max`) determines the maximum number of file handles that the Linux kernel allocates.
- The Oracle database background processes open all data files, logs, and other supporting files.
- The parameter must be high enough to include all the data files within your database and all supporting files.
- Set the kernel parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:
`fs.file-max = 6815744`
- View the setting in:

```
# cat /proc/sys/fs/file-max  
6815744
```

Asynchronous I/O (AIO)

- AIO is the kernel subsystem used to ensure that Oracle databases run properly on Linux.
- AIO allows a process to initiate several I/O operations without having to block or wait for any to complete.
- The process can retrieve the results of the I/O later.
- Set the maximum number of allowable concurrent requests kernel parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:
`fs.aio-max-nr = 1048576`
- View the setting in:

```
# cat /proc/sys/fs/aio-max-nr
1048576
```

Oracle-Related Shell Limits

- Three shell limits must be set for the `oracle` user:
 - `nofile`: Number of open file descriptors
 - `nproc`: Number of processes available to a single user
 - `stack`: Size of the stack segment of the process
- Soft limit versus hard limit
 - A hard limit can be changed only by `root`.
 - A soft limit can be changed by the user, up to the value of the hard limit.
- Define limits in the `/etc/security/limits.conf` file.
- OR edit `/etc/pam.d/login` file.
- A user can change a soft limit by using the `ulimit` command, for example:

```
$ ulimit -Sn 50
```

Add the following to the `/etc/security/limits.conf` file

- `oracle soft nproc 16384`
- `oracle hard nproc 16384`
- `oracle soft nofile 1024`
- `oracle hard nofile 65536`
- `oracle soft stack 10240`
- `oracle hard stack 32768`

HugePages

- HugePage sizes vary from 2 MB to 256 MB, based on the kernel version and the hardware architecture
 - Allow larger pages to manage memory
 - Are crucial for **faster Oracle database performance**
 - Are useful in both 32- and 64-bit configurations
 - Are integrated into the Linux kernel with release 2.6
 - Have been back-ported to some 2.4 kernels (2.4.21), but are implemented differently
 - Decrease page table overhead
 - Provide **faster overall memory performance**
 - Must be reserved during system startup
 - Are not swappable—there is no page-in/page-out overhead
- .

Configuring HugePages

- Guidelines exist for different OS versions and hardware architectures.
- Configuring HugePages on 64-bit Linux:
 - Set the `memlock` user limit in the appropriate file in `/etc/security/limits.d` slightly smaller than installed RAM.
 - Disable AMM by setting `MEMORY_TARGET` and `MEMORY_MAX_TARGET` to zero.
 - Use the `hugepages_settings.sh` script to calculate the recommended value for the `vm.nr_hugepages` parameter.
 - Edit `/etc/sysctl.d/97-oracle-database-sysctl.conf` and set the `vm.nr_hugepages` parameter.
 - Reboot your system.
- To check: `grep HugePages /proc/meminfo`

Oracle Database Smart Flash Cache (DBSFC)

- DBSFC:
 - Is available for both Oracle Solaris and Oracle Linux customers with the 11g R2, 12c, and 18c databases
 - Allows you to extend the Oracle Buffer Cache in memory (SGA) using secondary flash-based storage
 - Helps with read-only/read-mostly workloads
- When a block gets modified, it is modified in the standard database buffer cache, written to disk and copied over into the flash cache.
- A subsequent read can then be from this fast storage instead of from the originating data files.
- See <https://www.oracle.com/technetwork/articles/servers-storage-admin/smart-flash-cache-oracle-perf-361527.html>.

Oracle ASM

- For stand-alone or Oracle RAC databases, you must configure ASM on shared disk that is accessible by all cluster nodes.
 - Creating Oracle Clusterware files on shared file systems is deprecated.
- ASM performs the functions of a volume manager and a file system.
- ASM consists of a specialized Oracle instance and a set of disk groups that are managed through the ASM instance.
- A disk group is a set of disk devices that ASM manages.
 - Each disk device can be a partition, a logical volume, a RAID array, or a single disk.
 - ASM spreads data evenly across the disk group to optimize performance and usage.

ASM Library Driver (ASMLib)

- ASMLib simplifies the management of ASM disks.
- ASMLib has three components:
 - `oracleasm-support`: Provides user space shell scripts and is included with the Oracle Linux distribution
 - `oracleasmlib`: Provides the user space library and is installed from Unbreakable Linux Network (ULN)
 - `oracleasm`: Is the kernel driver included in `kernel-uek`

- To configure ASMLib:

```
# oracleasm configure -i
```

- To mark disks as ASM disks:

```
# oracleasm createdisk ASM_DISK_NAME candidate_disk
```

Using ASMLib Commands

Some options for the `oracleasm` program:

- `configure`: Configure the ASMLib driver. Options:
 - `-i` - interactive mode; `-e` - enable ASMLib startup on boot; `-d` - disable ASMLib startup on boot.
- `init`: Load and initialize the ASMLib driver.
- `exit`: Stop the ASMLib driver.
- `createdisk`: Mark a disk device for use with ASM.
- `deletedisk`: Unmark a named disk device.
- `querydisk`: Determine whether a disk device or disk name is being used by the ASMLib.
- `listdisks`: List the disk names of marked disks.
- `scandisks`: Check block devices for ASM disks.

ASM Filter Driver (ASMFD)

- Available on Linux systems starting with Oracle Database 12c Release 1
- Is an alternative to ASMLib
- Prevents accidental corruption or deletion of ASM devices
- Provides persistent paths and permissions for storage devices used with ASM
 - No need to update `udev` files with storage device paths and permissions.
- Cannot be run concurrently with ASMLib
 - If ASMLib is installed, it must be deinstalled before running ASMFD.

Quiz



Which of the following statements is true regarding ASM?

- a. ASMLib is required to use ASM.
- b. The `oracleasm` kernel driver is included in the Red Hat Compatible Kernel (RHCK).
- c. When using ASMLib, for Oracle Universal Installer (OUI) to recognize partitions as Oracle ASM disk candidates, you must mark the disk partitions that Oracle ASM can use.
- d. A RAID array cannot be included in an ASM disk group.



Quiz



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Summary

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Practices for Lesson 17: Overview

This practice covers the following topics:

- Using `scp` to upload `oracle` packages
- Installing and running Oracle Database Pre-Install
- Preparing disks for ASM use
- Installing and configuring ASMLib
- Reverting changes made to host03